

An Order-Zero Obstruction for the Davies–Helffer–Sjöstrand Calculus

Candidate full solution packet for arXiv:math/0212093

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Status: candidate full solution, likely valid; human review requested. For the natural algebraic interpretation of Remark 1.4 in Claire’s paper, the strict endpoint is necessary. We construct a spectrally positive closed operator satisfying the paper’s strongest resolvent exponent $\alpha = 0$ for which no bounded-operator algebra calculus on all of $S^0(\mathbb{R})$ can extend the established calculus on $\mathcal{A} = \bigcup_{\beta < 0} S^\beta$.

1 Source question

The source is Narinder S. Claire, *Functional Calculus for Semi-Bounded Operators* [1], arXiv: math/0212093. Its standing hypothesis is that H is closed and densely defined on a Banach space, $\sigma(H) \subset \mathbb{R}$, and for some $\alpha \geq 0$,

$$\|(z - H)^{-1}\| \leq c |\operatorname{Im} z|^{-1} \left(\frac{\langle z \rangle}{|\operatorname{Im} z|} \right)^\alpha, \quad z \notin \mathbb{R}. \quad (1)$$

For $\beta \in \mathbb{R}$, the symbol class S^β consists of the smooth functions g such that

$$|g^{(k)}(x)| \leq C_k \langle x \rangle^{\beta-k} \quad (k = 0, 1, \dots). \quad (2)$$

The paper builds a multiplicative bounded-operator calculus on $\mathcal{A} = \bigcup_{\beta < 0} S^\beta$ and says in Remark 1.4 (page 2):

“The strict inequality $\beta < 0$ is of importance and we conjecture that it cannot be extended to $\beta \leq 0$.”

Since $S^\beta \subset S^0$ for $\beta < 0$, the proposed endpoint is all of S^0 . Constants alone do not settle the conjecture: they can be adjoined by unitalization. We therefore use the following strong and natural formulation.

Endpoint extension problem. Must every operator satisfying (1) admit an algebra homomorphism

$$\Gamma : S^0(\mathbb{R}) \longrightarrow \mathcal{B}(X)$$

whose restriction to $\mathcal{A}$ is the Davies–Helffer–Sjöstrand calculus?

We introduce our main concepts.

1.1 Algebra of Slow Decreasing functions

Definition 1.1 $\langle z \rangle := (1 + |z|^2)^{\frac{1}{2}}$ for all complex z

Definition 1.2 For $\beta \in \mathbf{R}$ let S^β to be the set of all complex-valued smooth functions defined on $\mathbf{R}$ such that for every n there is a positive constant c_n where

$$\left| \frac{d^n f}{dx^n} \right| \leq c_n \langle x \rangle^{\beta-n}$$

Definition 1.3 We define the Algebra $\mathcal{A}$ as,

$$\mathcal{A} := \bigcup_{\beta < 0} S^\beta \tag{1.3}$$

Remark 1.4 The strict inequality $\beta < 0$ is of importance and we conjecture that it cannot be extended to $\beta \leq 0$.

Figure 1: The definition of $\mathcal{A}$ and Remark 1.4 in the source PDF, page 2.

2 Proof intuition

Smooth order-zero symbols may oscillate indefinitely on a logarithmic scale. We encode exactly those oscillations into a conditional Schauder basis. On the summing basis $s_n = e_1 + \cdots + e_n$ of c_0 , the diagonal operator $Hs_n = e^n s_n$ has an exceptionally good resolvent: the coefficients $(z - e^n)^{-1}$ have total variation at most $\pi/|\operatorname{Im} z|$. Thus H satisfies the source hypothesis already with $\alpha = 0$.

On the other hand, a smooth order-zero function can satisfy $f(e^n) = (-1)^n$. If an extension Γ existed, compactly supported spectral localizers and multiplicativity would force $\Gamma(f)s_n = (-1)^n s_n$. That is the alternating-sign multiplier on the summing basis, which is unbounded. The obstruction therefore concerns every possible multiplicative extension, not just convergence of the integral formula used in the source.

3 The result

Theorem 1 (Order-zero endpoint obstruction). *There exist a complex Banach space X and a closed densely defined operator H on X such that*

$$\sigma(H) = \{e^n : n \geq 1\} \subset (0, \infty)$$

and

$$\|(z - H)^{-1}\| \leq \frac{2\pi}{|\operatorname{Im} z|}, \quad z \notin \mathbb{R}. \tag{3}$$

Nevertheless, there is no algebra homomorphism

$$\Gamma : S^0(\mathbb{R}) \longrightarrow \mathcal{B}(X)$$

that agrees on $\mathcal{A} = \bigcup_{\beta < 0} S^\beta$ with the Davies–Helffer–Sjöstrand functional calculus of H .

More explicitly, the failure is witnessed by a real-valued $f \in S^0(\mathbb{R})$. No continuity assumption on Γ is required.

Corollary 2. *The analogous universal extension statement from $\mathcal{A}^+$ to S_+^0 for the semibounded calculus in Claire's paper is false as well.*

4 Proof

4.1 The summing basis and a variation estimate

Let $X = c_0$ with its usual norm and standard unit vectors (e_k) . Put

$$s_n = e_1 + \cdots + e_n \quad (n \geq 1).$$

This is the summing Schauder basis. Indeed, if $x = (x_k) \in c_0$ and $a_n = x_n - x_{n+1}$, then

$$x = \sum_{n=1}^{\infty} a_n s_n.$$

The N th basis projection $P_N x = \sum_{n \leq N} a_n s_n$ has coordinates

$$(P_N x)_k = \begin{cases} x_k - x_{N+1}, & k \leq N, \\ 0, & k > N, \end{cases}$$

so $P_N x \rightarrow x$ and

$$K := \sup_N \|P_N\| \leq 2. \quad (4)$$

We record the elementary multiplier estimate used below.

Lemma 3 (Bounded-variation multiplier). *Let (u_n) be a Schauder basis with basis constant K . If $c_n \rightarrow 0$ and*

$$V(c) := \sum_{n=1}^{\infty} |c_n - c_{n+1}| < \infty,$$

then the diagonal rule $M_c u_n = c_n u_n$ extends to a bounded operator and $\|M_c\| \leq KV(c)$.

Proof. Writing P_n for the basis projections, Abel summation gives

$$M_c x = \sum_{n=1}^{\infty} (c_n - c_{n+1}) P_n x.$$

The series converges absolutely in operator norm and has norm at most $KV(c)$. Its value on every basis vector is $c_n u_n$, which proves the claim. $\square$

4.2 A closed multiplier with positive spectrum

Let $\lambda_n = e^n$ and define the Schauder multiplier

$$D(H) = \left\{ x = \sum_{n \geq 1} a_n s_n : \sum_{n \geq 1} \lambda_n a_n s_n \text{ converges in } c_0 \right\},$$

$$Hx = \sum_{n \geq 1} \lambda_n a_n s_n.$$

The finite linear span of the s_n lies in $D(H)$, hence H is densely defined. It is closed: if $x^{(j)} \rightarrow x$ and $Hx^{(j)} \rightarrow y$, continuity of the basis coefficient functionals gives

$$a_n(y) = \lambda_n a_n(x) \quad (n \geq 1).$$

The Schauder expansion of y therefore shows that the defining series for Hx converges to y .

Fix $z = x + iy \notin \mathbb{R}$ and set $c_n = (z - e^n)^{-1}$. The chord joining two successive reciprocal values is no longer than the intervening curve, so

$$\begin{aligned} \sum_{n=1}^{\infty} |c_n - c_{n+1}| &\leq \sum_{n=1}^{\infty} \int_{e^n}^{e^{n+1}} \frac{dt}{|z - t|^2} \\ &= \int_e^{\infty} \frac{dt}{(t - x)^2 + y^2} \leq \frac{\pi}{|y|}. \end{aligned} \quad (5)$$

Lemma 3 and (4) give a bounded diagonal operator $R_z = M_c$ satisfying (3). Moreover,

$$e^n c_n = z c_n - 1.$$

Consequently $R_z x \in D(H)$ and $(z - H)R_z x = x$. Conversely, comparison of basis coefficients gives $R_z(z - H)u = u$ for $u \in D(H)$. Thus $R_z = (z - H)^{-1}$.

The same argument identifies the entire spectrum. For real $t \notin \{e^n : n \geq 1\}$, the sequence $(t - e^n)^{-1}$ tends to zero and has finite total variation: after finitely many terms it is monotone, and the remaining finite initial segment causes no problem. The preceding inverse argument therefore shows $t \in \rho(H)$. Each e^n is an eigenvalue with eigenvector s_n . Hence

$$\sigma(H) = \{e^n : n \geq 1\}.$$

This also verifies the source hypothesis with its strongest allowed exponent $\alpha = 0$.

4.3 One order-zero symbol encodes the bad signs

Choose a real $\chi \in C^\infty(\mathbb{R})$ with $\chi = 0$ on $(-\infty, 1]$ and $\chi = 1$ on $[e, \infty)$. Define

$$f(t) = \begin{cases} \chi(t) \cos(\pi \log t), & t > 0, \\ 0, & t \leq 0. \end{cases} \quad (6)$$

This is smooth because it vanishes on a neighborhood of the origin. For $t \geq e$, repeated differentiation expresses $f^{(k)}(t)$ as t^{-k} times a fixed linear combination of $\sin(\pi \log t)$ and $\cos(\pi \log t)$. On the compact transition interval $[1, e]$ all derivatives are bounded. Therefore

$$|f^{(k)}(t)| \leq C_k \langle t \rangle^{-k},$$

so $f \in S^0(\mathbb{R})$. Its spectral values are

$$f(e^n) = \cos(\pi n) = (-1)^n. \quad (7)$$

4.4 Multiplicativity forces an unbounded operator

Suppose, toward a contradiction, that $\Gamma : S^0(\mathbb{R}) \rightarrow \mathcal{B}(c_0)$ is an algebra homomorphism extending the existing calculus on $\mathcal{A}$. For each n choose $\phi_n \in C_c^\infty(\mathbb{R})$ such that $\phi_n(e^n) = 1$ and whose support contains no other e^j . Both ϕ_n and $f\phi_n$ belong to $C_c^\infty(\mathbb{R}) \subset \mathcal{A}$.

The source calculus has the usual eigenvector identity

$$g(H)s_j = g(e^j)s_j \quad (g \in C_c^\infty(\mathbb{R})). \quad (8)$$

For completeness, applying its almost-analytic integral to s_j replaces $(z - H)^{-1}s_j$ by $(z - e^j)^{-1}s_j$; the scalar Cauchy–Green identity then gives (8). Agreement and multiplicativity now imply

$$\Gamma(f)\phi_n(H) = \Gamma(f)\Gamma(\phi_n) = \Gamma(f\phi_n) = (f\phi_n)(H).$$

Applying this to s_n and using (8) gives

$$\Gamma(f)s_n = f(e^n)s_n = (-1)^n s_n. \quad (9)$$

But the multiplier in (9) is not bounded on the summing basis. Indeed, let

$$x_N = \sum_{n=1}^N (-1)^n s_n.$$

Every coordinate of x_N is an alternating tail sum, so $\|x_N\|_\infty = 1$. Equation (9) would give

$$\Gamma(f)x_N = \sum_{n=1}^N s_n, \quad \|\Gamma(f)x_N\|_\infty = N,$$

contradicting $\Gamma(f) \in \mathcal{B}(c_0)$. This proves Theorem 1.

Finally, restrict f and the compactly supported localizers to $[0, \infty)$. Claire’s semibounded calculus agrees with the whole-line calculus after smooth extension and has the same eigenvector identities. The identical contradiction proves Corollary 2.

5 Verification notes and limitations

The proof is exact and uses no numerical input. The main review checkpoints are: (i) the Abel-summation multiplier estimate; (ii) closedness and the two-sided inverse on the stated multiplier domain; (iii) the logarithmic derivative bounds in (6); and (iv) the localization step forcing (9).

The result resolves the source conjecture in the universal extension sense stated above. It does *not* assert that every individual order-zero symbol is obstructed, nor that no proper subalgebra of S^0 can be adjoined. In particular, adjoining constants is harmless. It also does not claim that Claire supplied a more formal endpoint statement than Remark 1.4; the packet makes the tested formulation explicit.

Novelty check. On 11 August 2026 we searched the run’s registry, solution, attempt, and proof-gap indexes; the exact paper title and author; the quoted endpoint phrase; and combinations of “Davies–Helffer–Sjöstrand S^0 ”, “conditional Schauder basis”, “summing basis”, and “slow decreasing functional calculus”. The search found the source, later general work on smooth calculi and spectral mapping, and standard Schauder-multiplier techniques, but no later resolution of this exact conjecture or this construction. Novelty confidence is therefore moderate, not definitive.

Human-review recommendation. Check as a candidate full solution, with particular attention to whether the intended meaning of the source’s concise Remark 1.4 is exactly the universal algebra-extension formulation made explicit here. Under that formulation, the proof rules out the endpoint even for discrete positive spectrum and the strongest resolvent exponent.

References

- [1] N. S. Claire, *Functional Calculus for Semi-Bounded Operators*, arXiv:math/0212093 (2002).
- [2] E. B. Davies, *The Functional Calculus*, J. London Math. Soc. (2) **52** (1995), 166–176.